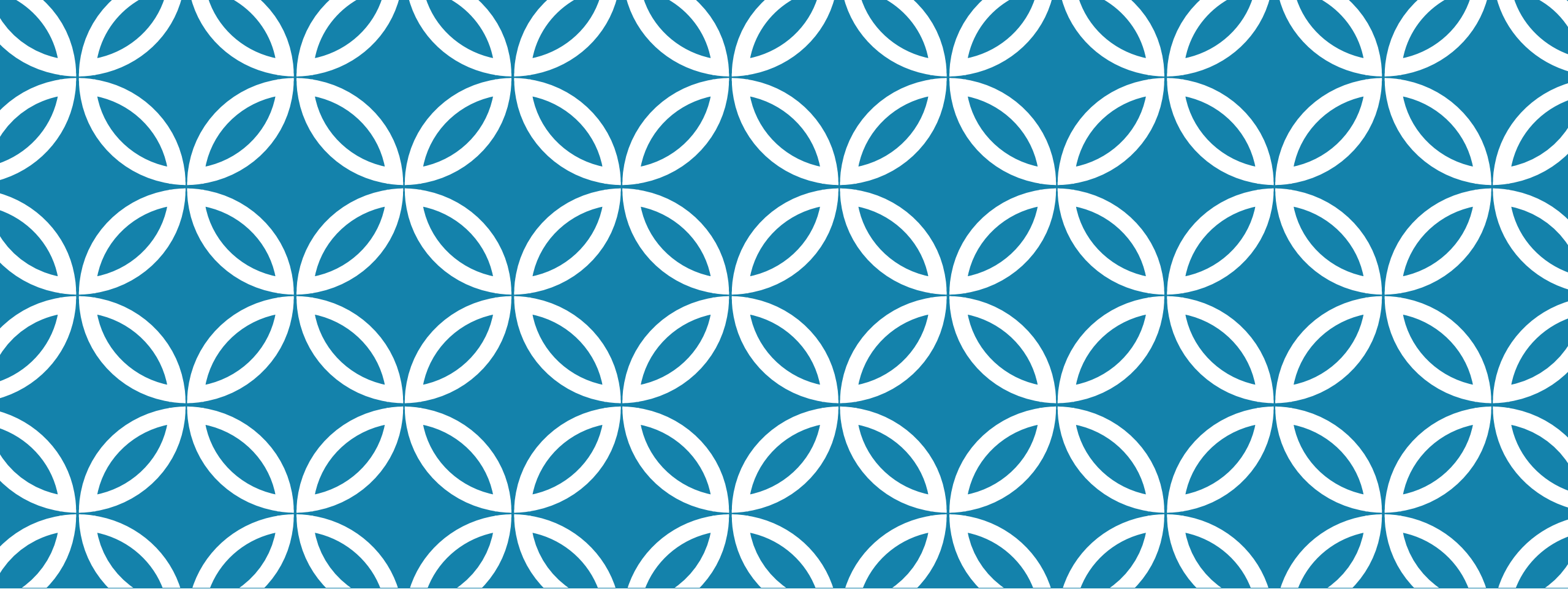




FUZZY LOGIC & FUZZY SETS

Data Mining

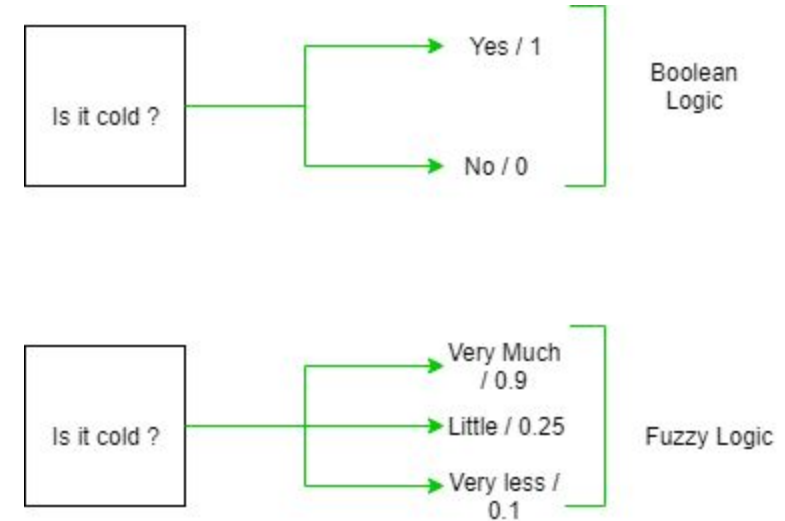
WHAT IS FUZZY LOGIC?

The term **fuzzy** refers to things that are not clear or are vague. In the real world many times we encounter a situation when we can't determine whether the state is true or false, their fuzzy logic provides very valuable flexibility for reasoning. In this way, we can consider the inaccuracies and uncertainties of any situation.

In the boolean system truth value, 1.0 represents the absolute truth value and 0.0 represents the absolute false value. But in the fuzzy system, there is no logic for the absolute truth and absolute false value. But in fuzzy logic, there is an intermediate value too present which is partially true and partially false.

INTRODUCTION

Computers mainly use Boolean Logic to determine the result of scenarios. As per the Boolean Logic, value 1 refers to True, and 0 means False. The term Fuzzy refers to something unclear or vague. The computer cannot easily understand such cases. Thus, it cannot produce an exact result of True or False. But a Fuzzy Logic algorithm makes systems more intelligent and helps them understand the problems where there may be other answers than true or false.



ADVANTAGES OF FUZZY LOGIC SYSTEM

This system can work with any type of inputs whether it is imprecise, distorted or noisy input information.

The construction of Fuzzy Logic Systems is easy and understandable.

Fuzzy logic comes with mathematical concepts of set theory and the reasoning of that is quite simple.

It provides a very efficient solution to complex problems in all fields of life as it resembles human reasoning and decision-making.

The algorithms can be described with little data, so little memory is required.

DISADVANTAGES OF FUZZY LOGIC SYSTEMS

Many researchers proposed different ways to solve a given problem through fuzzy logic which leads to ambiguity. There is no systematic approach to solve a given problem through fuzzy logic.

Proof of its characteristics is difficult or impossible in most cases because every time we do not get a mathematical description of our approach.

As fuzzy logic works on precise as well as imprecise data so most of the time accuracy is compromised

APPLICATION OF FUZZY LOGIC SYSTEMS

It is used in the aerospace field for altitude control of spacecraft and satellites.

It has been used in the automotive system for speed control, traffic control.

It is used for decision-making support systems and personal evaluation in the large company business.

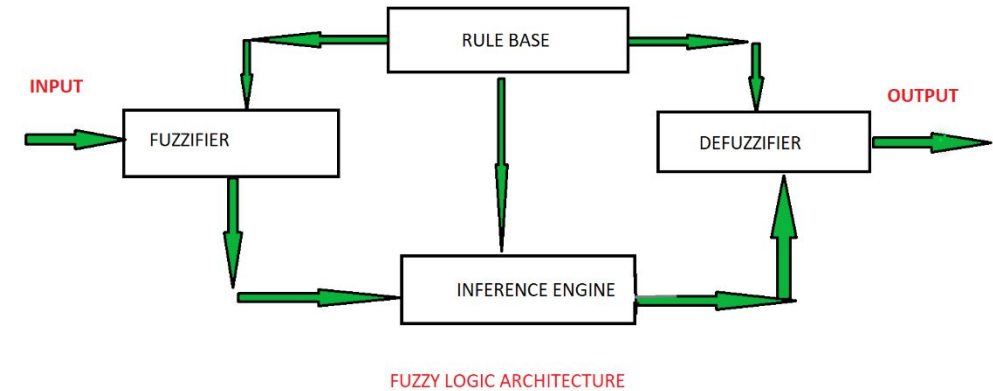
It has application in the chemical industry for controlling the pH, drying, chemical distillation process.

Fuzzy logic is used in Natural language processing and various intensive applications in Artificial Intelligence.

Fuzzy logic is extensively used in modern control systems such as expert systems.

Fuzzy Logic is used with Neural Networks as it mimics how a person would make decisions, only much faster. It is done by Aggregation of data and changing it into more meaningful data by forming partial truths as Fuzzy sets.

ARCHITECTURE OF FUZZY LOGIC SYST



Its Architecture contains four parts :

RULE BASE: It contains the set of rules and the IF-THEN conditions provided by the experts to govern the decision-making system, on the basis of linguistic information. Recent developments in fuzzy theory offer several effective methods for the design and tuning of fuzzy controllers. Most of these developments reduce the number of fuzzy rules.

FUZZIFICATION: It is used to convert inputs i.e. crisp numbers into fuzzy sets. Crisp inputs are basically the exact inputs measured by sensors and passed into the control system for processing, such as temperature, pressure, rpm's, etc.

INFERENCE ENGINE: It determines the matching degree of the current fuzzy input with respect to each rule and decides which rules are to be fired according to the input field. Next, the fired rules are combined to form the control actions.

DEFUZZIFICATION: It is used to convert the fuzzy sets obtained by the inference engine into a crisp value. There are several defuzzification methods available and the best-suited one is used with a specific expert system to reduce the error.

APPLICATIONS OF FUZZY LOGIC

- In automobiles, fuzzy logic is used for gear selection and is based on factors such as engine load, road conditions and style of driving.
- In dishwashers, fuzzy logic is used to determine the washing strategy and power needed, which is based on factors such as the number of dishes and the level of food residue on the dishes.
- In copy machines, fuzzy logic is used to adjust drum voltage based on factors such as humidity, picture density and temperature.
- In aerospace, fuzzy logic is used to manage altitude control for satellites and spacecrafts based on environmental factors.
- In medicine, fuzzy logic is used for computer-aided diagnoses, based on factors such as symptoms and medical history.
- In chemical distillation, fuzzy logic is used to control pH and temperature variables.
- In natural language processing, fuzzy logic is used to determine semantic relations between concepts represented by words and other linguistic variables.
- In environmental control systems, such as air conditioners and heaters, fuzzy logic determines output based on factors such as current temperature and target temperature.